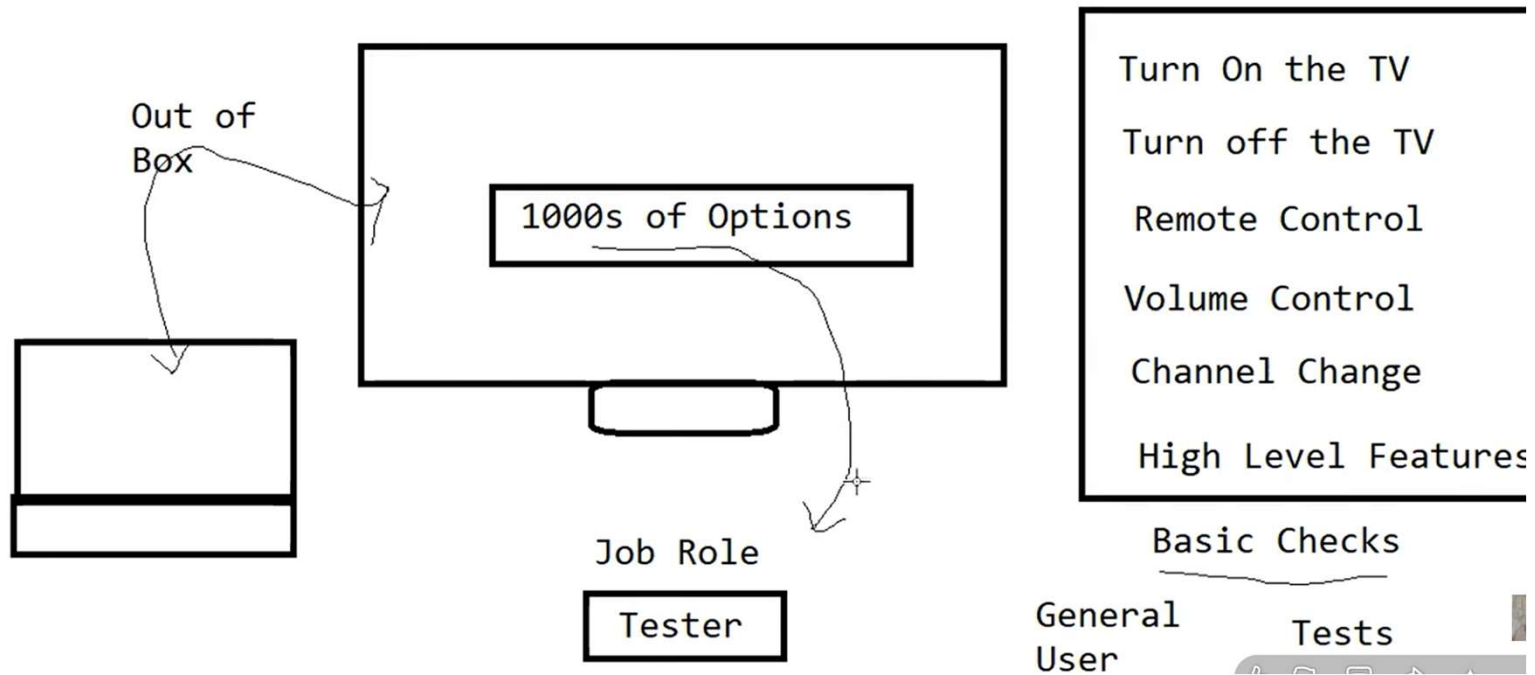


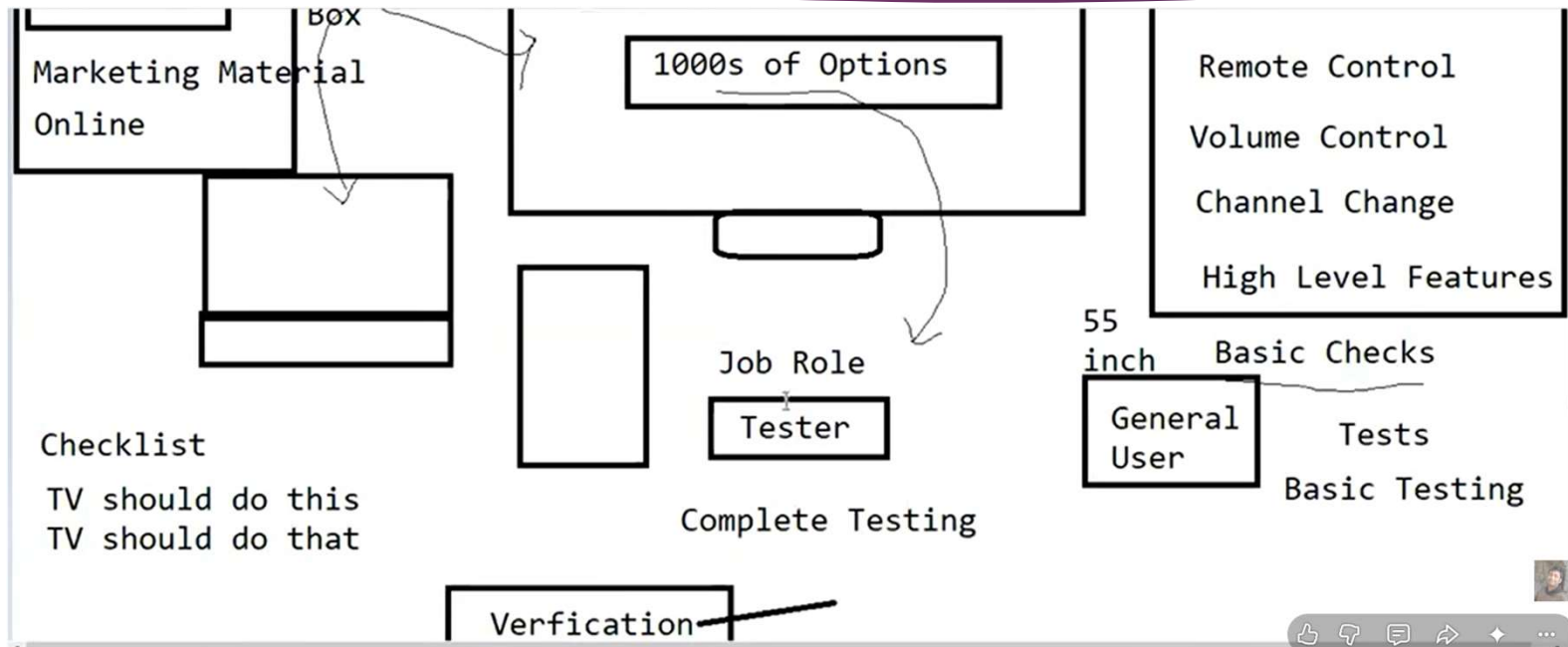
SOFTWARE TESTING –QUALITY ASSURANCE



Introduction Testing



Testing



Testing

Validation

Checklist

- C1 Turn on the TV using TV Button
- C2 Turn on the TV using Remote Button
- C3 Turn off the TV using TV Button
- C4 Connect a Laptop to TV using HDMI cables
- C5 Connect a Laptop to TV using XYZ cables

Test Coverage

99.99%

Testing

TV should do this
TV should do that

Complete Testing

basic testing

Verfication
Validation

Verification Testing
(Requirements)

Checklist

- C1 Turn on the TV using TV Button PASS
- C2 Turn on the TV using Remote Button Fail
- C3 Turn off the TV using TV Button
- C4 Connect a Laptop to TV using HDMI cables
- C5 Connect a Laptop to TV using XYZ cables

Good

Test Coverage

Testing

TV should do this
TV should do that

Complete Testing

Verification
Validation

Verification Testing
(Requirements)

Validation Testing (End User is
satisfied)

Checklist

- C1 Turn on the TV using TV Button PASS
- C2 Turn on the TV using Remote Button Fail
- C3 Turn off the TV using TV Button
- C4 Connect a Laptop to TV using HDMI cables
- C5 Connect a Laptop to TV using XYZ cables

Test Coverage

end User

5 mins





What is Testing ?

- ▶ **Testing** means the process of checking, trying out, or evaluating a person, product, or system to find out how it works, what it knows, or if it has any faults

What is SoftWare ?

- ▶ Set of Programs to solve a purpose.
- ▶ MS Word, MS Paint, WhatsApp, Axis Mobile App.

Software is a set of **programs, instructions, and data** that tells a computer or electronic device **what to do**.

In simple words:

Software = Instructions that make hardware work.



What is SoftWare Testing ?

- ▶ SoftWare Testing is the process of verification and validation of the software to check whether it is working as per the expectations.
- ▶ **Software testing is the systematic process of evaluating and verifying that a software application functions correctly, securely, and efficiently according to its specified requirements.** Its primary goal is to identify bugs, defects, or gaps between expected and actual results before the software is released to end-users.

Purpose of SoftWare Testing ?

- ▶ The primary purpose of software testing is to **verify that a software application works correctly, meets specified requirements, and is free of critical defects before reaching the end-user**. It acts as a quality control gate that ensures reliability, security, and optimal performance.
- ▶ Positive Intention of improving the quality of the SoftWare.
 - **Bug Discovery:** Finding errors, defects, and flaws in the code during development.
 - **Requirement Validation:** Verifying that the product behaves exactly as outlined in design documents.
 - **Defect Prevention:** Learning from discovered bugs to optimize future coding processes.
 - **Quality Assurance:** Guaranteeing that the product meets both technical and UX standards

Why Software Testing is Required ?

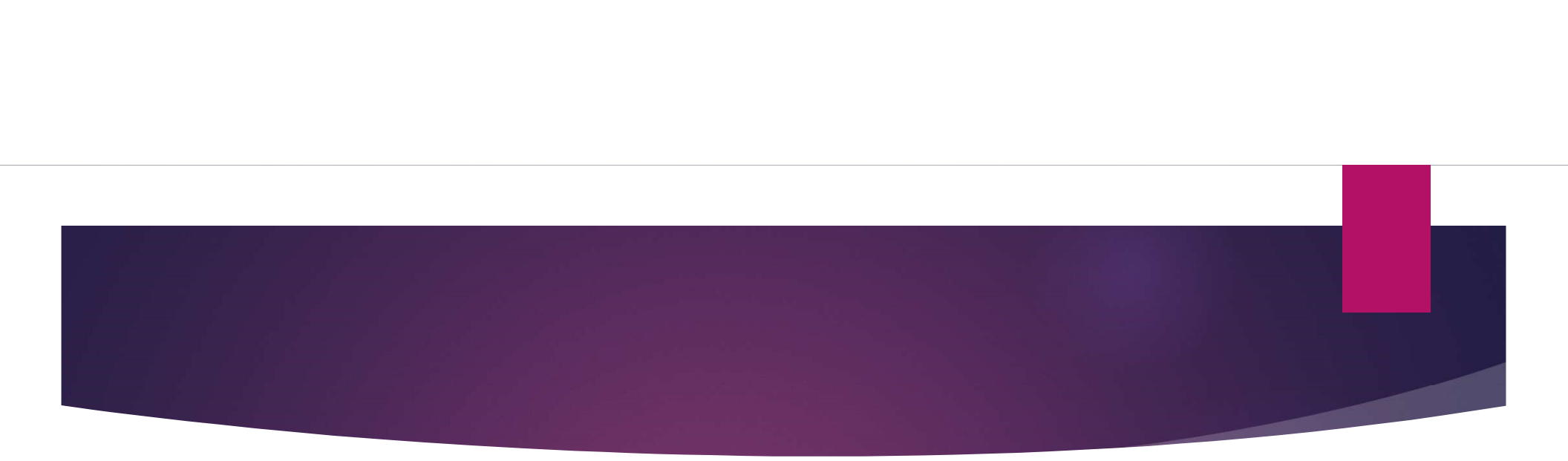
► The below are few things that will happen if you don't perform testing:

1. Buggy software with less quality may get released into the market
2. Business Reputation will be lost
3. Customer may file a case in court
4. Sales may get dropped
5. Revenue loss
6. Loss of life
7. And many more may happen.



Practicing Software Testing on this demo application

- ▶ Understanding the Application Functionality.
- ▶ Practice Creating Test Plan
- ▶ Practice Creating Test Scenarios.
- ▶ Practice Creating Test Cases
- ▶ Finding defects in this Application and reporting defects



client will be sharing the application

client want us to test

step 1 understand the application

step 2 create test plan

step 3 create Test Scenarios

step 4 create Test cases

step 5 Finding Defects and
report them

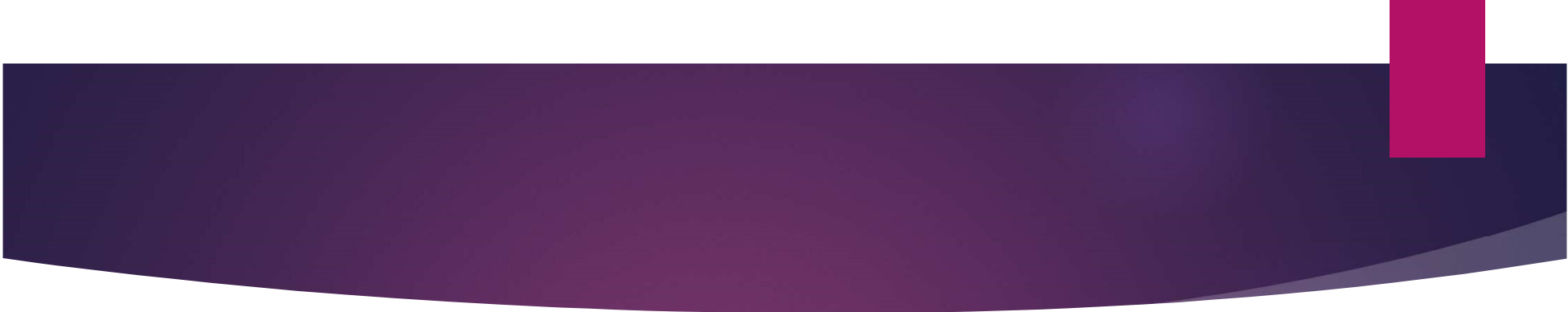


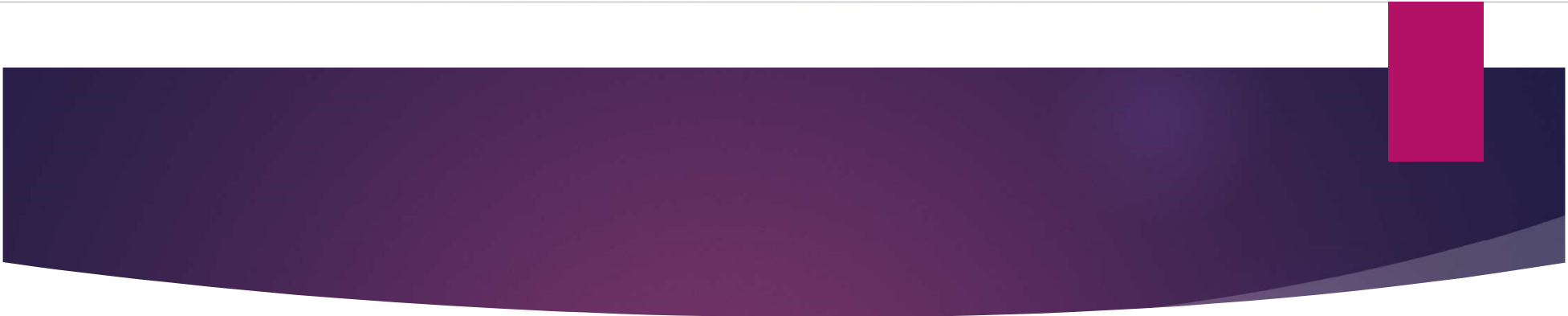
Test Artifacts



Domain knowledge plays a key role in software Testing

- ▶ As Software Testers, we need to have good domain knowledge of the Application we are testing.
- ▶ Example Application: <http://tutorialsninja.com/demo>
- ▶ What is the domain of the above example application?
- ▶ There are several domains:
 - Amazon.com - e-Commerce Domain
 - hdfcbank.com - Banking Domain

- 
- <https://www.hdfcergo.com/> - Insurance Domain
 - https://www.youtube.com/watch?v=QDnU1q64vww&feature=emb_title - Health-care Domain -
<https://www.adroitinfosystems.com/products/ehospital-systems>
 - <https://www.orangehrm.com/> - ERP Domain (Enterprise Resource Planning)
 - Google Class Room or <https://www.schoology.com/> - Education Domain
 - And there are several other domains that are not mentioned here



Search Functionality

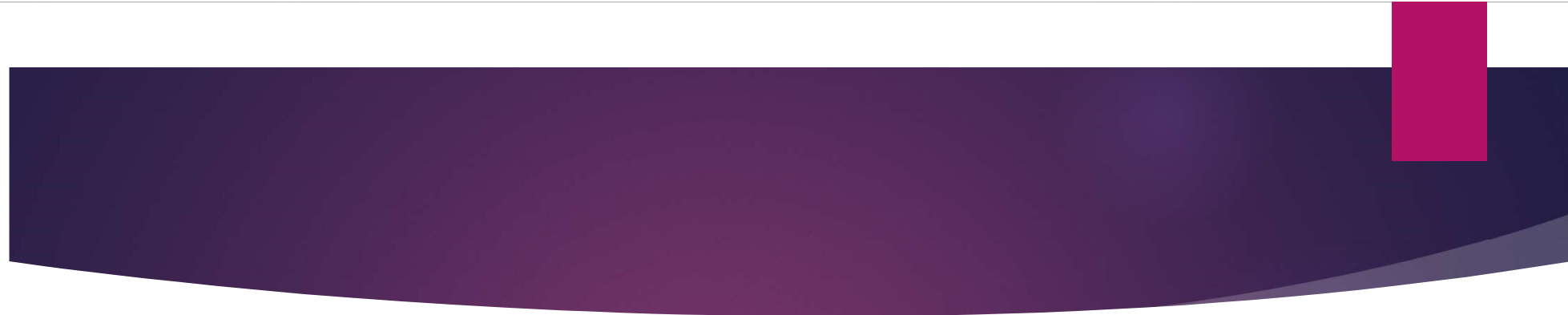
Product Display Page

eCommerce Domain

Wish List

Wedding Registry





eCommerce Domain

Wish List

Wedding Registry

1	Dinning Set	Dining Table	1
10	Spoons	Cooker	1
		Refrigirator	1
15	Cups	TV	2

Bride and Groom

Guest

eCommerce Domain
+

Search Functionality

Product Display Page

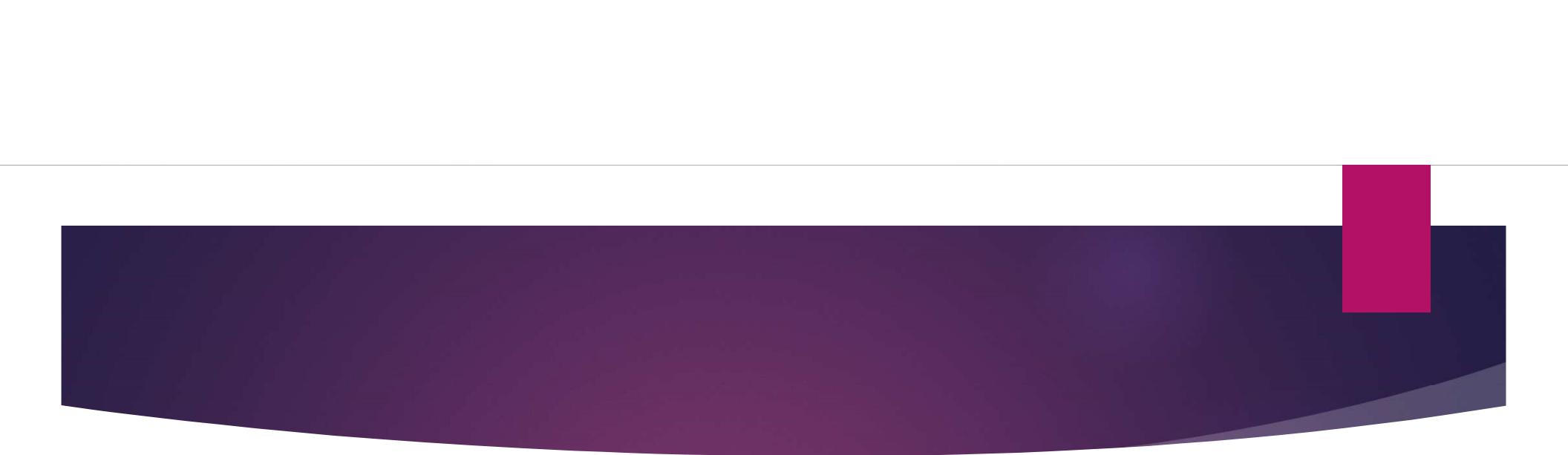
Wish List

Wedding Registry

Bride and Groom

Guest

1	Dinning Set	Dining Table	1
10	Spoons	Cooker	1
		Refrirator	1



eCommerce - Amazon.com

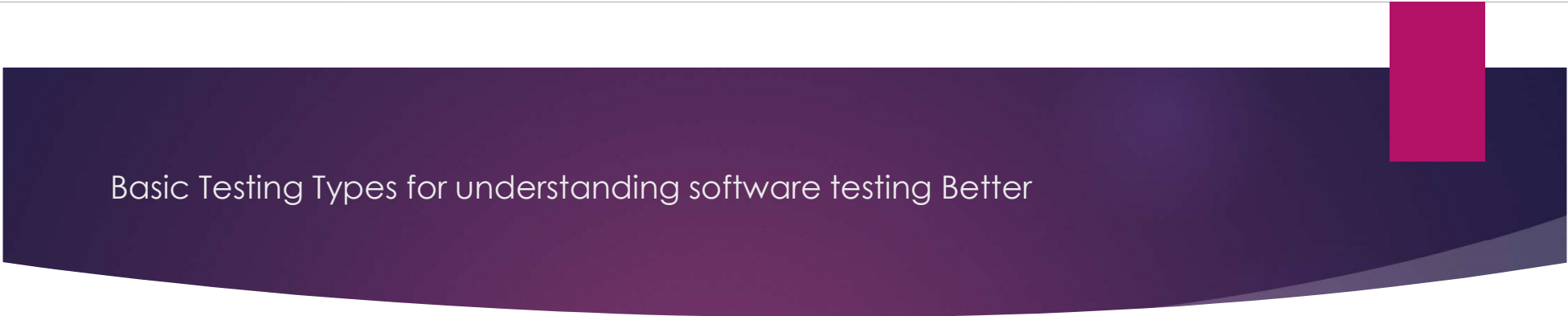
Banking - hdfcbank.com

Insurance - hdfcergo.com

Healthcare - <https://www.adroitinfosystems.com/products/ehospital-systems>

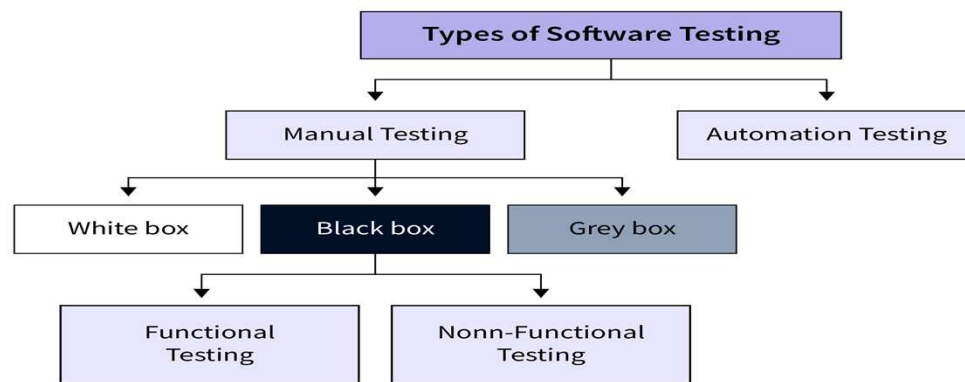
ERP Domain - orangehrm.com

Education Domain - google classrooms



Basic Testing Types for understanding software testing Better

- ▶ We need to understand the below basic Testing Types for understanding Software Testing better:
 - Manual Testing
 - Automation Testing
 - Manual Testing
 - Automation Testing
 - UI Testing
 - API Testing
 - Database Testing



SoftWare Testing Career Planning

► The below is the plan for a better Software Testing Career:

1. Start with learning of **Manual Testing**
2. Next go for **Selenium Java** Automation
3. Then learn **Appium** for automation Mobile Applications
4. Learn **UFT** for automating Desktop Applications
5. Learn API/Web Services Automation with **Rest Assured Framework**
6. Learn Automating Angular JS Applications with **Protractor** Tool

SoftWare Testing Career Planning

1. Learn **Performance Testing** using **JMeter** or LoadRunner
2. Learn **Security Testing** using BurpSuite tool and also manual way
3. Add-on Learnings
 1. Learn **Test Management** Tools like Test Rail or ALM etc.
 2. Learn **SQL** and **DB Testing**
 3. Learn **Linux Commands**
 4. Learn **DevOps**
 5. Learning **Agile** Methodologies and Practices
 6. Learn Future trends of Testing like **IOT, AI, Machine Learning** etc. in testing perspective
 7. Learning different popular Programming Languages like **Python, C#, JavaScript** etc.
 8. Learning latest Test Automation Tools
4. Never stop learning and continuously learn new things
5. Invest in Yourself

Software Testing Career Planning

Step#1

Manual Testing

Domain Knowledge

eCommerce
Healthcare
Banking
Insurance
Education
ERP

Step#2

Selenium and Java

Core Java

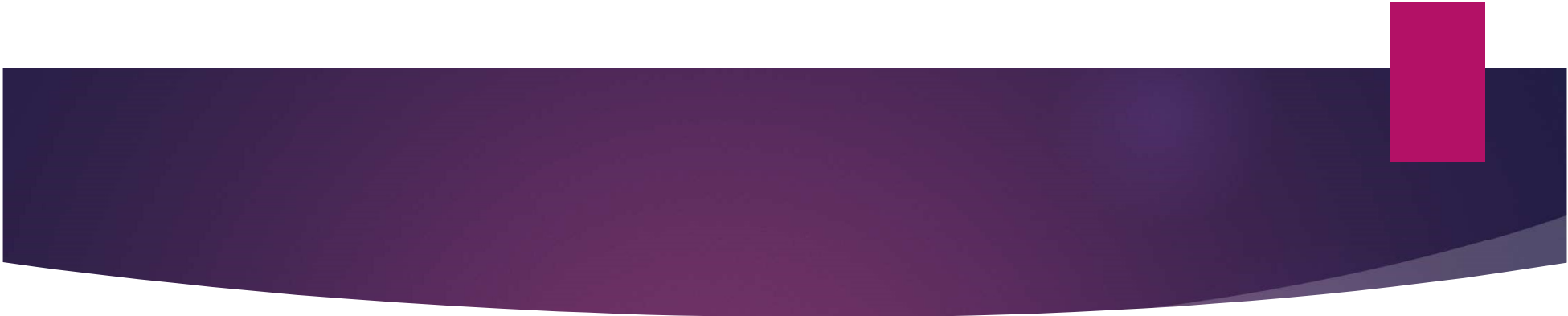
Web Applications
Automation

Step#3

Appium Automation Tool

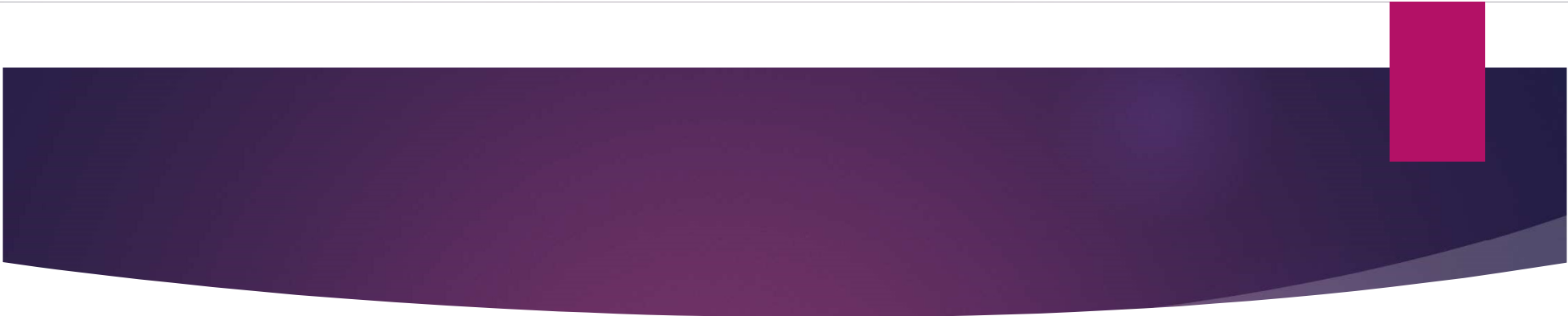
Mobile Applications
Automation





Step#4 **UFT** Automation Tool
Desktop Applications
Automation

Step#5 API / WebServices
Automation
API Testing
**Rest Assured
Framework**



Step#6

Protractor

Automation Tool

Angular JS
Applications

Step#7 Performance Testing

JMeter

LoadRunner



Step#8 Security Testing

Step#10 SQL
Database Testing

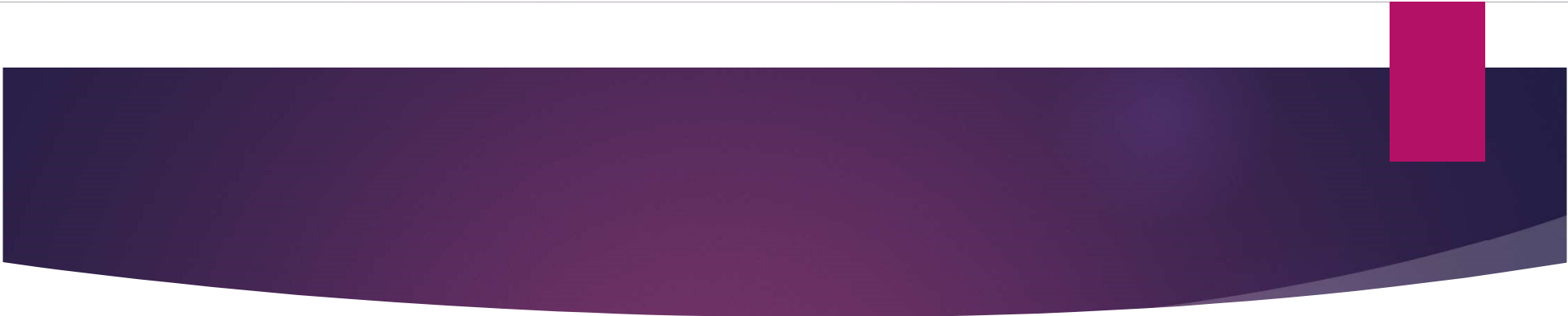
Step#12 DevOps

Step#9 Test Management
Tools

TestRail
ALM

Step#11 Linux Commands

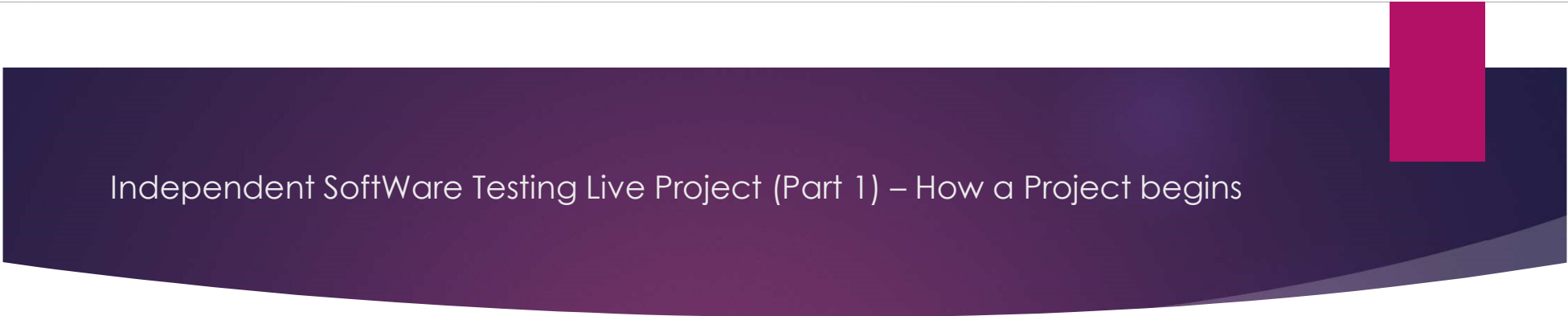
Step#13 Agile Methodoloes
and Practices



Invest in Yourself

Software Testing
Conferences

Rare courses

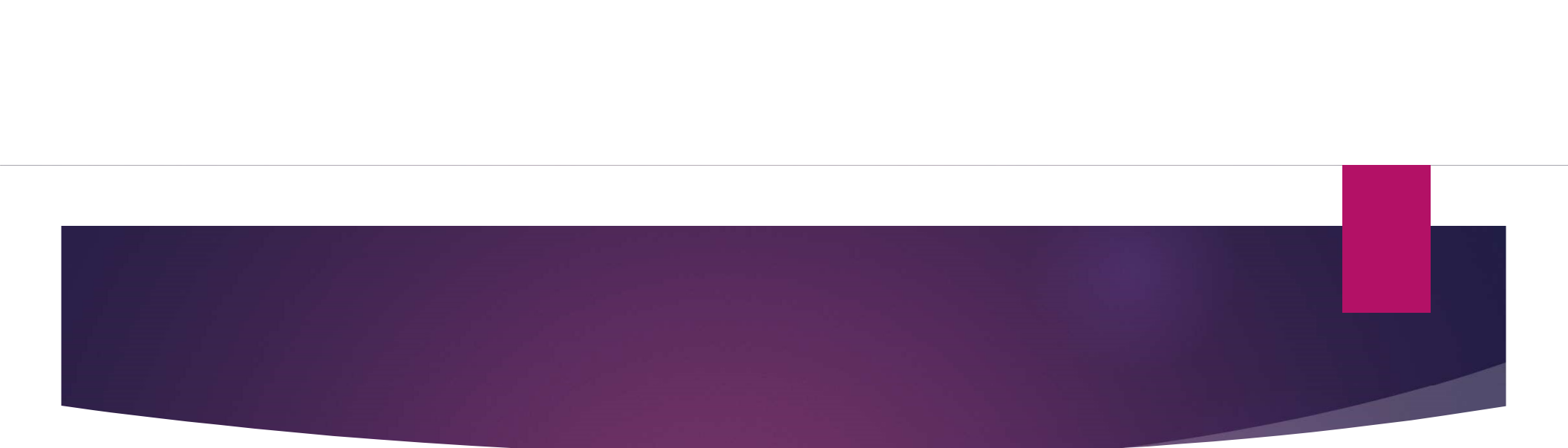


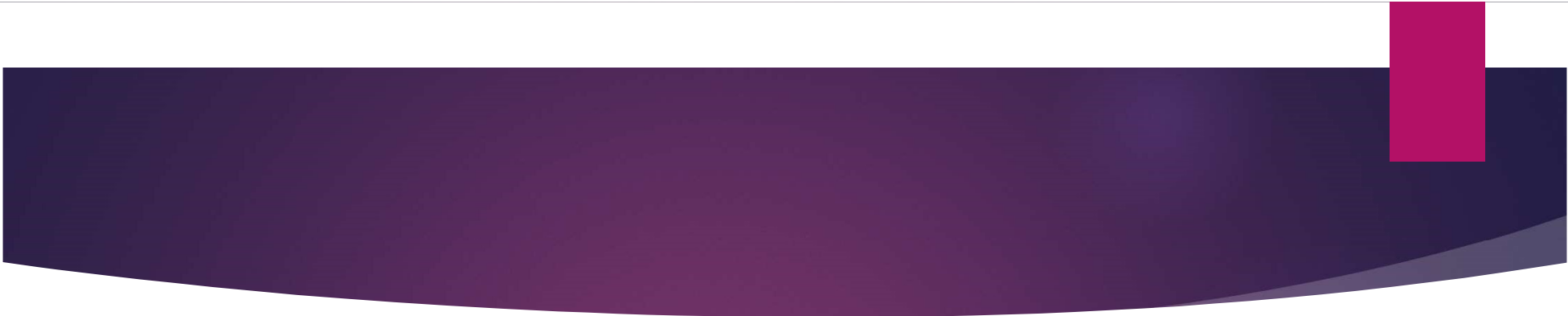
Independent SoftWare Testing Live Project (Part 1) – How a Project begins

1. BDE (Business Development Engineer) at the company will search for the Clients and emails them regarding the different Independent Testing Services provided by the company
2. Once he gets a reply/lead, he will arrange a kickoff meeting with the Client, Test Manager, Test Lead and himself
3. In Kickoff meeting, client gives the details about the Project and clearly mentions his requirements
4. Let's say the Client has given the <http://tutorialsninja.com/demo/> application and asked us to start exploring the application till we decided to meet again in a week
5. Testing team explores the application for a week

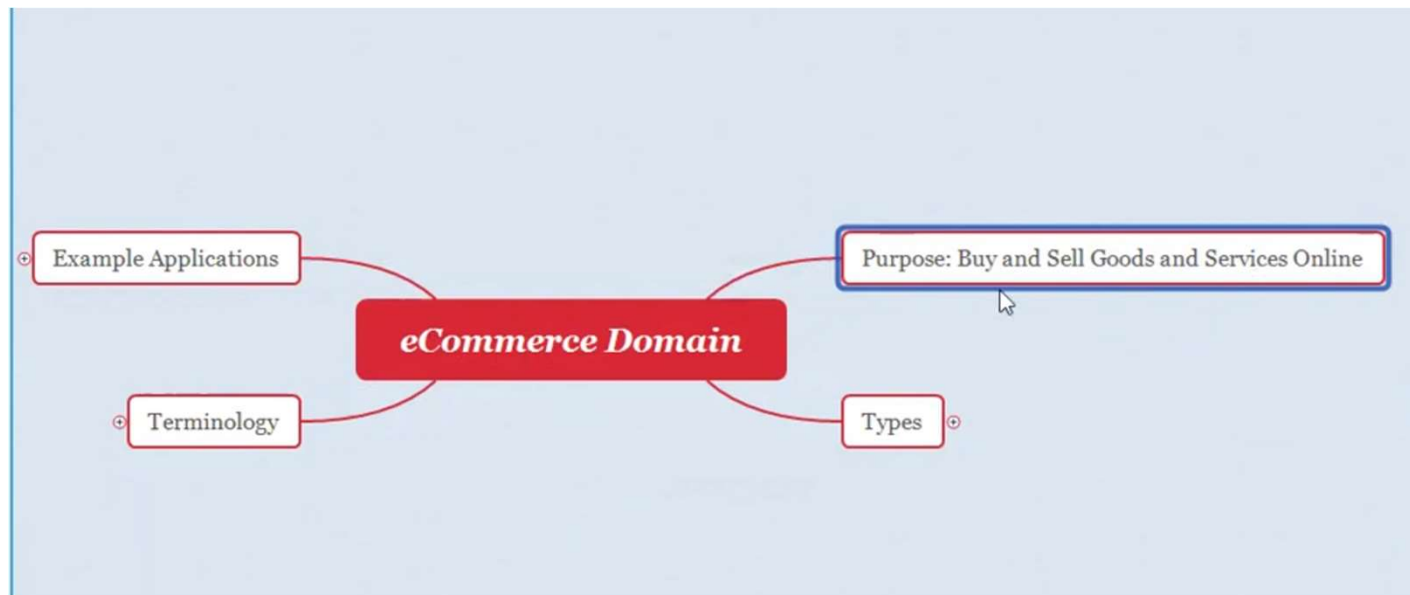
eCommerce Domain knowledge

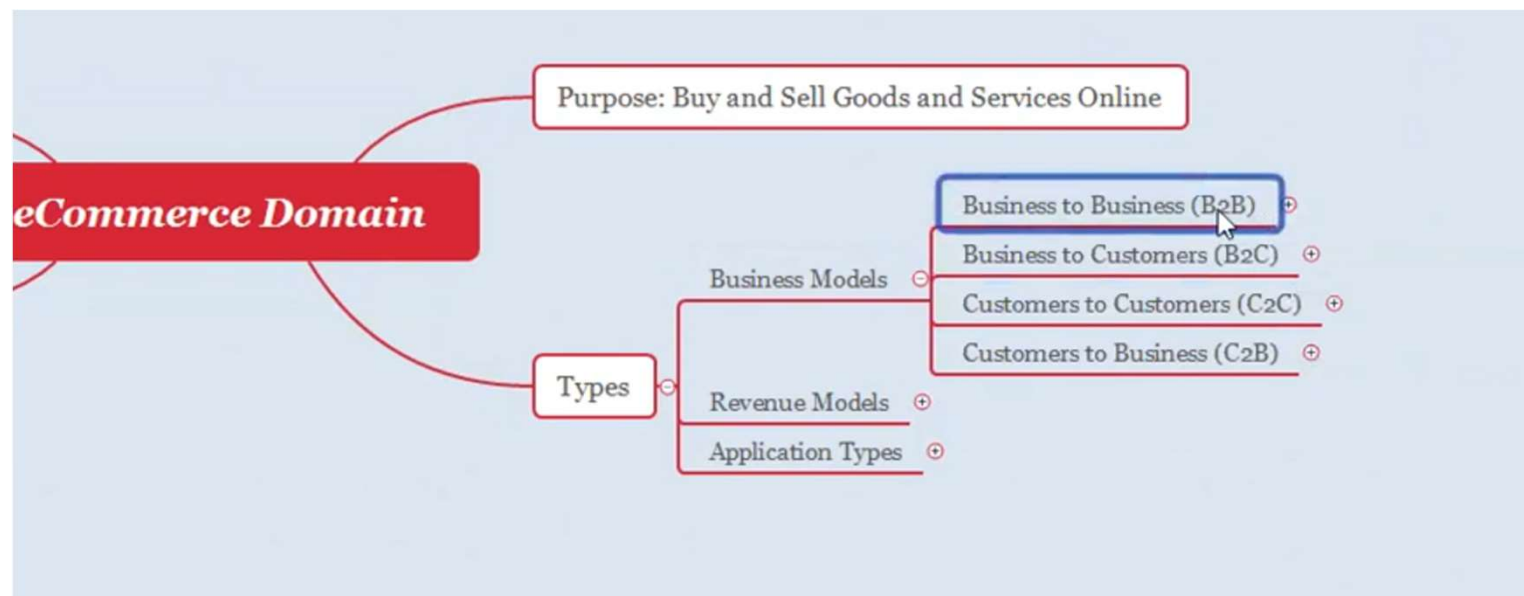
- Lets say the client has given an e-Commerce Application - <http://tutorialsninja.com/demo/> for testing
- We should have some eCommerce Domain knowledge to understand the Application very well.
- The below are the things which will help us in understanding the eCommerce Domain knowledge well:
- Purpose of eCommerce Applications Buy and Sell Goods and Services on-lineTypes of eCommerce ApplicationsCategorized based on Business ModelsBusiness to Business (B2B) - Automobile company purchasing its parts from other companiesBusiness to Customers (B2C) - <http://amazon.com/>Customers to Customers (C2C) - <https://www.olx.in/>Customers to Business (C2B) - StockPhotos - <https://submit.shutterstock.com/?language=en>Categorized based on Revenue

- 
- Models
 - Dropshipping - Amazon.com - Selling others products online
 - Subscription - Paying subscription fee for utilising the services - Course Learning Website
 - Private labelling and manufacturing - Selling your own products online - <https://order.karachibakery.com/>
 - Whitelabelling - Selling others owned products with their brand
 - Wholesaling - Middle man between manufacturer and retailer
 - Rent or loan - Renting a product instead of buying it
 - Freemium - Free to purchase model - <https://www.spotify.com/in/> - zoom online meetings
 - Categorized based on Application Types
 - Individual Brand Website - <https://shop.adidas.co.in/>
 - Online Retailers with selected number of Sellers - <https://www.macys.com/>
 - Marketplaces having multiple sellers - <https://www.amazon.in/>
 - Example eCommerce Applications
 - <https://www.amazon.com/>
 - <https://www.flipkart.com/>
 - <https://www.ebay.com/>
 - <https://www.groupon.com/>
 - https://www2.hm.com/en_in/index.html
 - <https://paytm.com/>
 - <https://www.starbucks.com/>
 - <https://www.trivago.in/>
 - <https://www.myntra.com/>
 - <https://www.uber.com/>
 - <https://www.walmart.com/>
 - <https://www.kohls.com/>
 - <https://www.target.com/>
 - <https://www.rossstores.com/>
 - <https://www.jcpenney.com/>
 - <https://www.makemytrip.com/>
 - <https://www.olx.in/>
 - <https://www.macys.com/>
 - <https://submit.shutterstock.com/?language=>

- 
- enAnd many moreTerminology:Product Display PageShopping CartGuest CheckoutSignIn CheckoutWedding RegistryWish ListAffiliate MarketingBuyers PersonaCall-To-ActionContact Management SystemCookiesmCommerceMerchant AccountEmail MarketingPayment GatewaysHero ImageSearch CategoriesFeatured ProductsRelated ProductsRecommended ProductsReturn ItemsTrack PackageOrder ManagementEverything is a terminologyAnd many more terms<https://www.outerboxdesign.com/ecommerce-website-design/ecommerce-glossary-terms><https://www.tychesoftwares.com/ecommerce-terms-glossary/><https://blog.smile.io/ecommerce-terminology-need-know/><https://webkul.com/blog/50-ecommerce-terms-you-should-know/><https://ecommerceguide.com/guides/ecommerce-glossary/><https://cm-commerce.com/academy/ecommerce-terms-glossary/>Google search for many more such to increase your knowledge

eCommerce domain knowledge





Domain

Types

Business Models

- Dropshipping ○ Amazon.com - Selling other products online
- Subscription ○ Paying subscription fee for utilising the services - Course Learning Website
- Private labelling and manufacturing ○ Selling your own products online ○ <https://order.>

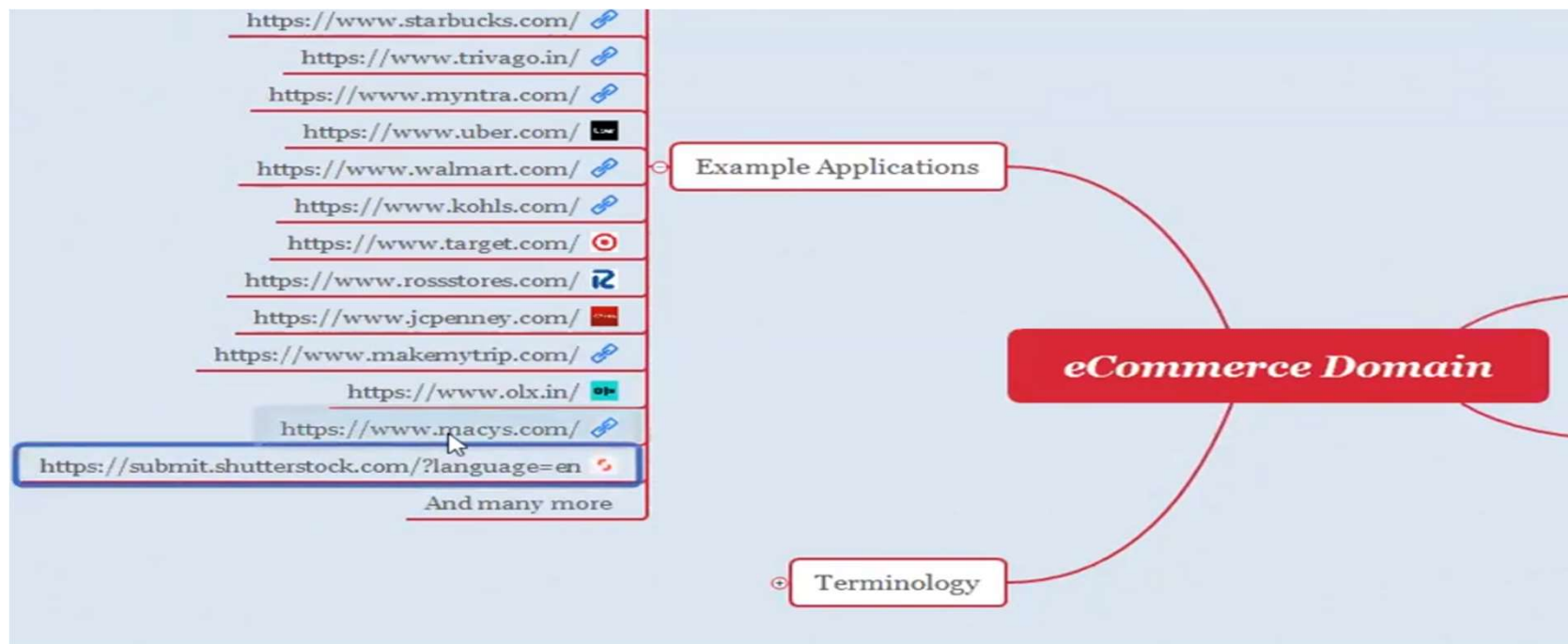
Revenue Models

- Whitelabelling ○ Selling other owned products with their brand
- Wholesaling ○ Middle man between manufacturer and retailer
- Rent or loan ○ Renting a product instead of buying it

- Freemium ○ Free to purchase model ○ <https://www.spotify.com/in/> 
zoom online meetings

Application Types





Product Display Page

Shopping Cart

Guest Checkout

SignIn Checkout

Wedding Registry

Wish List

Affiliate Marketing

Buyers Persona

Call-To-Action

Content Management System

Cookies

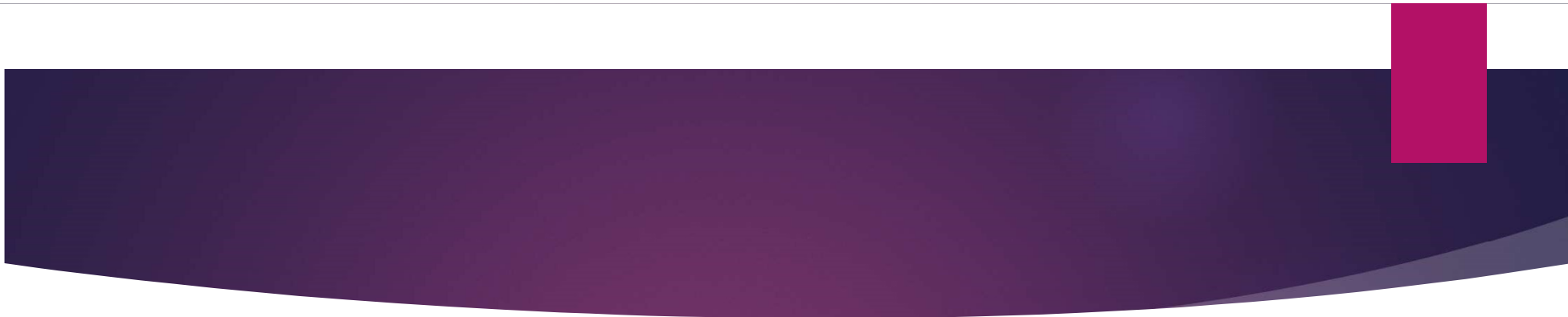
mCommerce

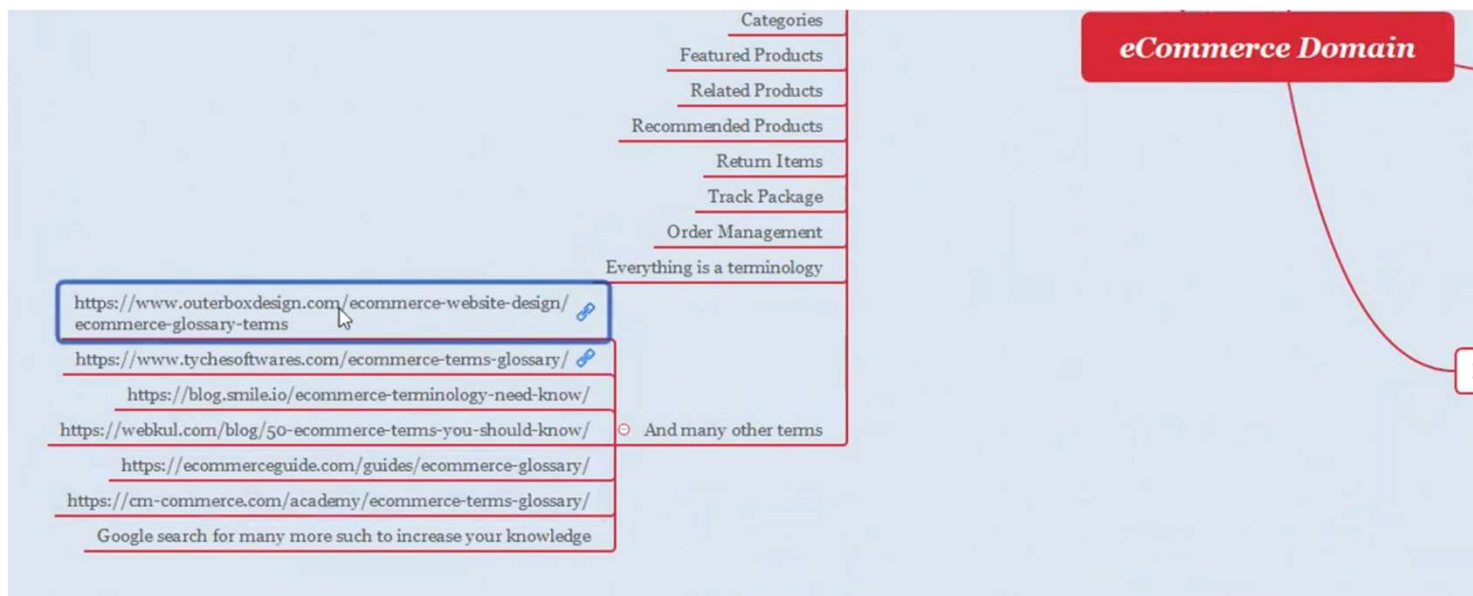
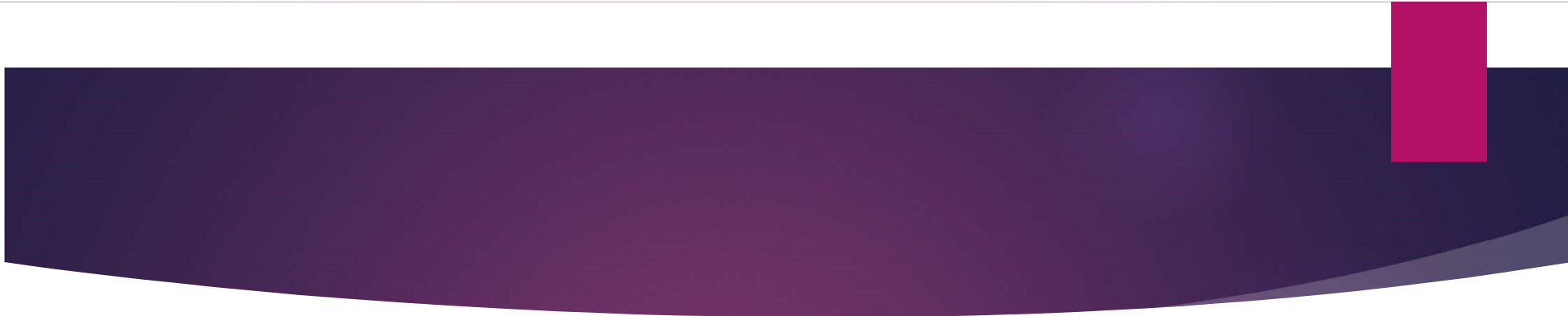
Merchant Account

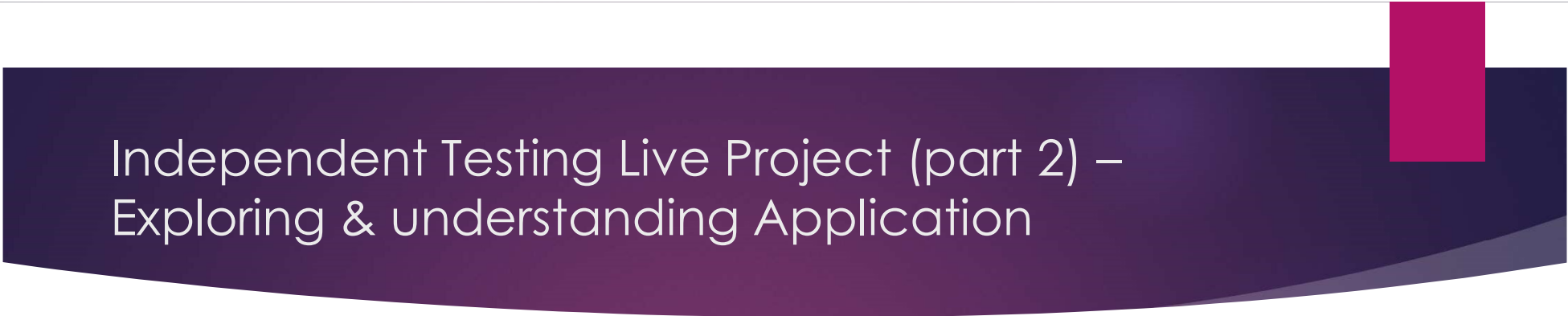
Cookies

mCommerce

Merchant Account







Independent Testing Live Project (part 2) – Exploring & understanding Application

- Session 7 - How a Project begins is explained (Part 1)
- Session 8 - eCommerce Domain Knowledge required for this Project is shared and explained
- In this session, we will go to the next step of the Project where we will explore and understand the Application.



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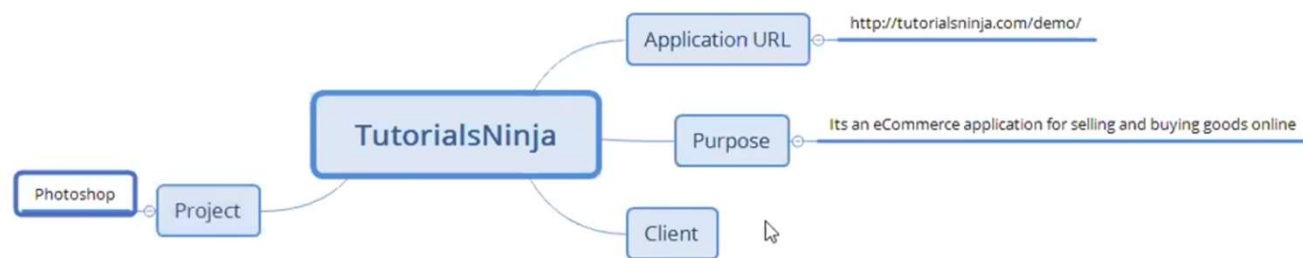
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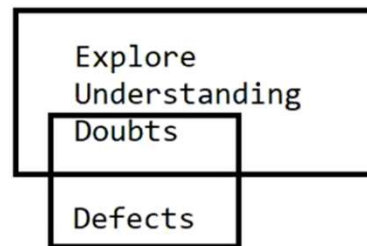
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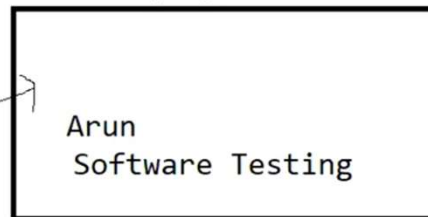


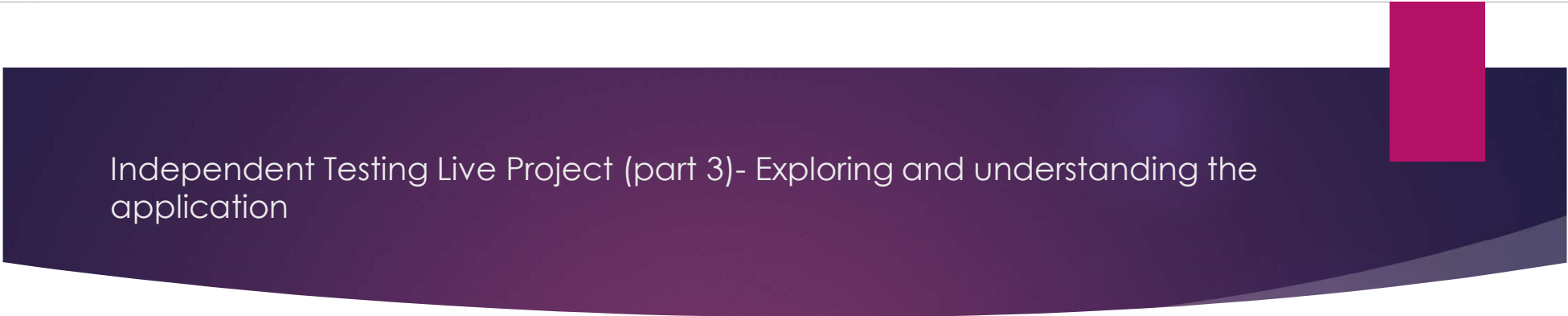
7 days

I
Client

Photoshop

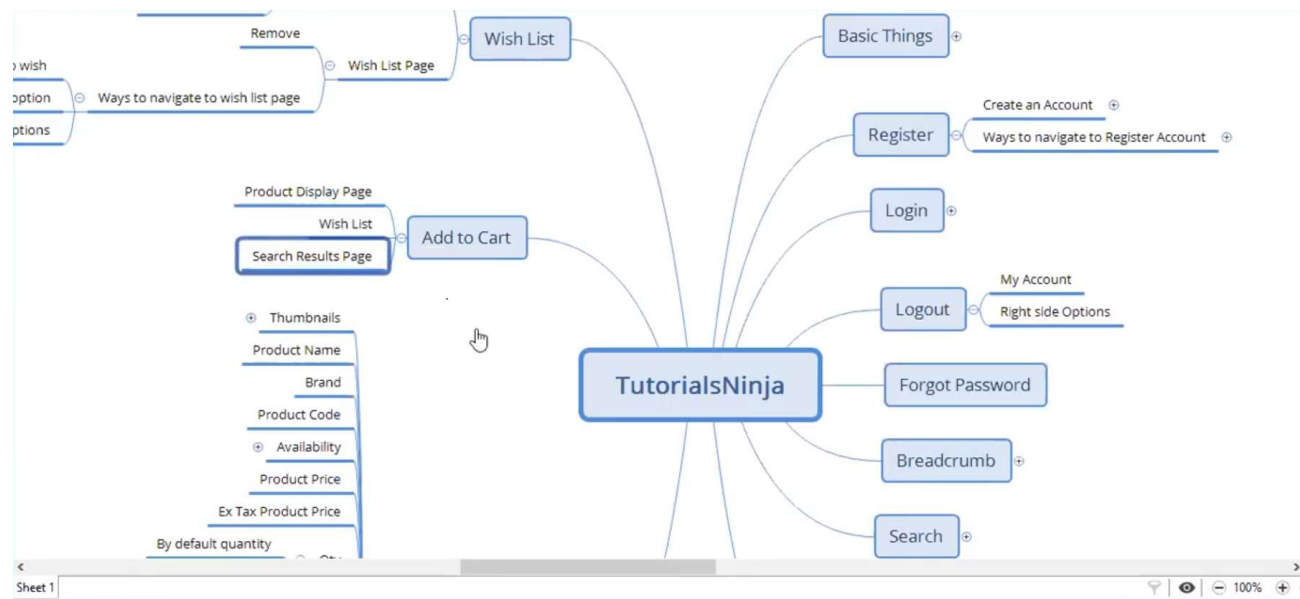
XYZ Company

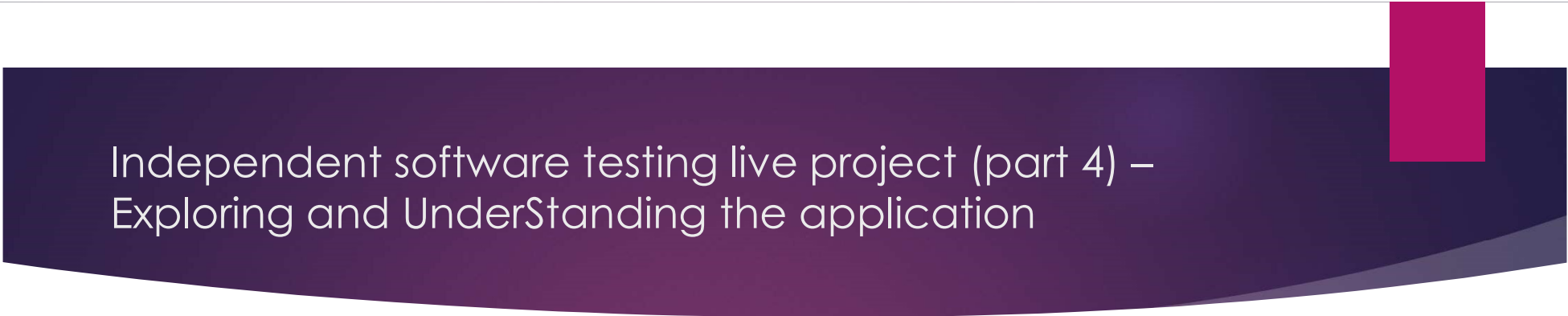




Independent Testing Live Project (part 3)- Exploring and understanding the application

- In Session 9, we have started exploring and understand the Application.
- In this Session, we will continue exploring and understanding the Application.





Independent software testing live project (part 4) – Exploring and UnderStanding the application

- In Session 9, we have started exploring and understand the Application.
- In Session 10, we will continue exploring and understanding the Application.
- In this Session, we will continue exploring and understanding the remaining of the Application.

Independent software testing live project (part 5) – test planning and test strategy

- After the Project Begins, we explore the application and clear our doubts which came while exploring the application
- After exploring the Project and clearing our doubts, we will start creating the Test Plan document
- Test Plan is a document required for planning the test activities for delivering the quality product.
- Test Plan ([Sample Test Plan](#))
- Test Plan is a document
Test Plan Contains: Overview/Objective
In Scope (Features to be Tested)
Out of Scope (Features not to be Tested)
Test Approach (i.e. Test Strategy)
Test Environments
Test Deliverables and Milestones
Test Resources
Test Schedules
Entry Criteria
Exit Criteria
Suspension and Resumption Criteria
Tools
Risks and Mitigations
Approvals
- [Finalized Test Plan](#)



Independent software testing live project (part 6) – Test scenarios

- Test Scenarios Template ([View Here](#))
- Finalized Test Scenarios Template ([View Here](#))